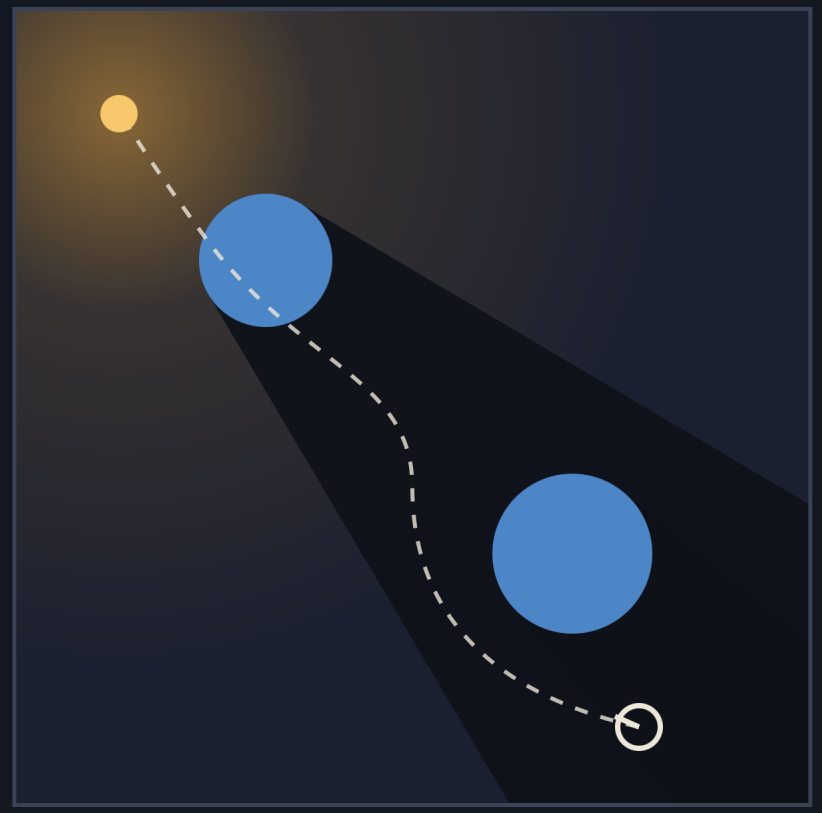
B.TECH PROJECT · IIIT-DELHI · 2026

Learning multiple tasks in parallel

A Horde of general value functions, learned off-policy, on a simulated robot in a 3 × 3 m arena with pillars and shadows.



GQ(λ) · tile coding · 7 demons · 12M steps

Knowledge as predictions: the general value function

$$v(s) = \mathrm{E}\pi\,[\, C_{t+1} + \gamma_{t+1} C_{t+2} + \gamma_{t+1}\gamma_{t+2} C_{t+3} + \cdots \mid S_t = s\,]$$

The expected sum of a signal C if the robot followed policy π, fading out at rate $1 - \gamma$. Change the four functions and the same learner answers a different question.

Horde (Sutton et al., 2011): many such demons learn in parallel from one stream of experience, each one off-policy, while the robot does something else entirely.

π

target policy

whose behaviour is imagined

$\gamma(s)$

termination

how far ahead to look

$c(s)$

cumulant

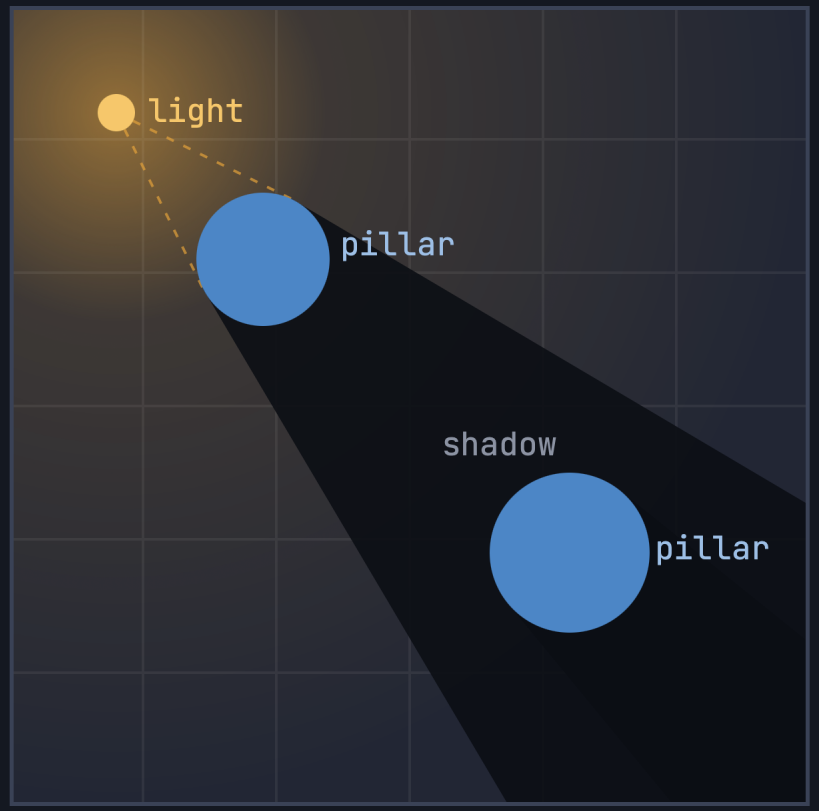
the signal being summed

$z(s)$

terminal value

added when it stops

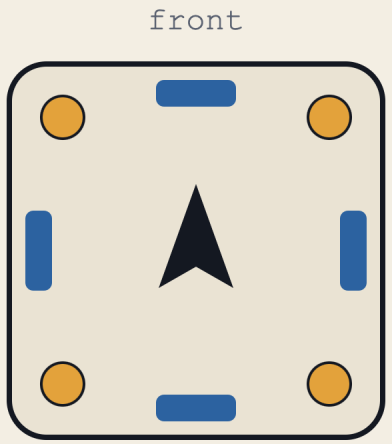
One light, two pillars, hard shadows



`3 × 3 m`	square arena; the robot keeps 0.12 m from walls
`(0.4, 2.6)`	the single light source, in one corner
`r 0.25, 0.30`	pillars at (0.95, 2.05) and (2.10, 0.95) m; the first blocks the straight path to the light
`32.6 %`	of the reachable floor is in shadow, where every light sensor reads zero

$$\text{dark} \Leftrightarrow \exists\, k{:}\ \|c_k - s_k u\| < r_k \text{ and } 0 < s_k < d,\ s_k = c_k \cdot u$$

Thirteen noisy sensors and five actions



infrared proximity, one per side

light sensor, one per corner

OBSERVATION · 13 NUMBERS

infrared ×4	front, left, right, back · higher when close
light ×4	front-left, front-right, back-left, back-right
motion ×2	speed, turn rate
bump ×1	contact switch, 0 or 1
compass ×2	cos and sin of the heading

ACTIONS · ONE EVERY 0.5 S

FORWARD · BACKWARD · LEFT · RIGHT · STOP

Tile coding: 54 of 14,735 features switch on

$$q(s, a) = \theta^{\mathsf{T}} \varphi(s, a) = \Sigma_i \, \theta_i$$

summed over the 54 active tiles of (s, a): 53 tiles + 1 bias, one copy per action (5 × 2,947)

Small overlapping groups of sensors are tiled jointly instead of one huge joint tiling, following the Horde paper's "overlapping joint tilings".

Step sizes are divided by 54, so one update moves $q(s, a)$ by $\alpha \cdot \delta$. All seven demons share the same φ.

sensor group	bins	tilings
IR front	16	8
IR left × IR right	8	4
light FL × FR	12	8
v	12	8
ω	8	4
IR front × v	8	4
IR front × ω	8	4
compass (cos θ, sin θ)	8	4
bump	2	1
IR front × IR back	8	4
light BL × BR	8	4

06 · LEARNING RULE

Every demon learns with GQ(λ), off-policy

$$\delta_t = C_{t+1} + (1 - \gamma_{t+1})\, z_{t+1} + \gamma_{t+1}\, \theta^T \bar{\varphi}_{t+1} - \theta^T \varphi_t$$

$$\bar{\varphi}_{t+1} = \Sigma_a\, \pi(a \mid S_{t+1})\, \varphi(S_{t+1}, a)$$

$$e_t = \varphi_t + \gamma_t\, \lambda\, \min(1, \rho_t)\, e_{t-1}$$

$$\theta \leftarrow \theta + \alpha\, [\, \delta_t e_t - \gamma_{t+1}(1 - \lambda)(w^T e_t)\, \bar{\varphi}_{t+1}\,]$$

$$w \leftarrow w + \beta\, [\, \delta_t e_t - (w^T \varphi_t)\, \varphi_t\,]$$

θ: value weights · w: secondary weights for the gradient correction · e: eligibility trace. Highlighted cap added at 10.9M steps (slide 13).

behaviour μ · sticky random

$$\mu(a) = 0.04 + 0.8 \cdot \mathbb{1}[a = A_{t-1}]$$

target π · softmax, τ = 0.1

$$\pi(a|s) \propto \exp(q(s, a) / \tau)$$

importance ratio

$$\rho_t = \pi(A_t|S_t) / \mu(A_t) \in [0, 25]$$

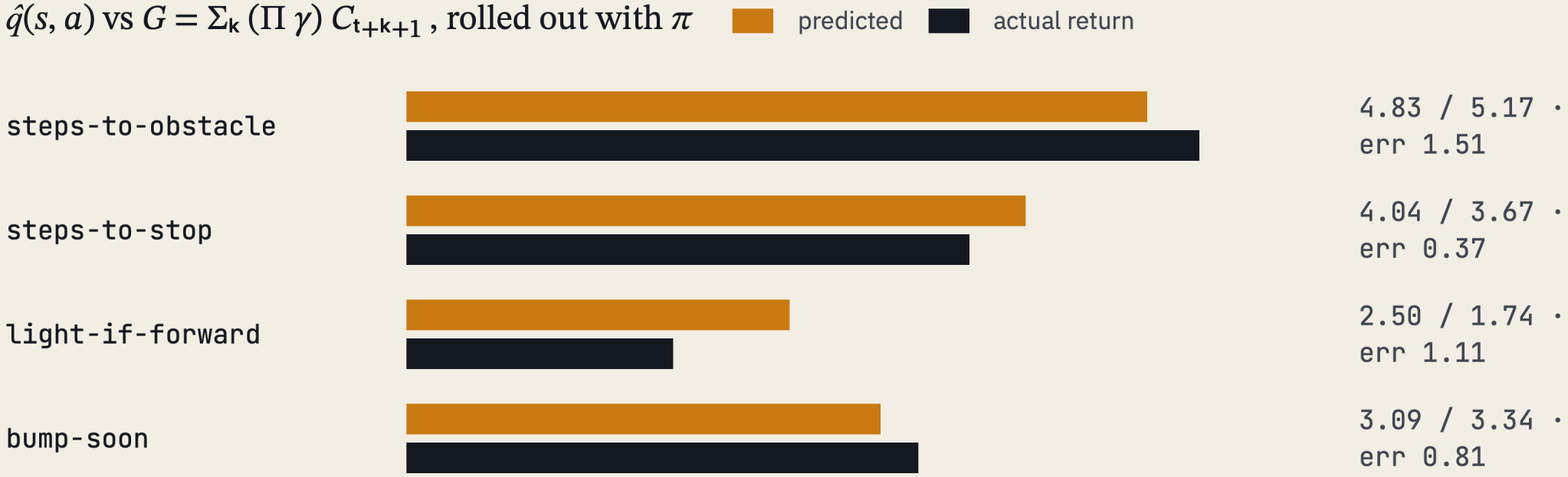
Seven demons, one stream of experience

Four **prediction** demons ask “what if I kept doing X?”. Three **control** demons learn the greedy policy for their own cumulant.

demon	π	cumulant c	γ	λ	α
steps-to-obstacle	FORWARD	1	0 at obstacle or bump	0.4	0.3
steps-to-stop	STOP	1	0 once |v| < 0.03	0.4	0.3
light-if-forward	FORWARD	$\Sigma_j L_j$	0.9	0.4	0.2
bump-soon	FORWARD	bump	0.8	0.4	0.2
seek-light	softmax(q)	L_FL + L_FR	0.95	0.3	0.1
spin	softmax(q)	|ω| / 3	0.95	0.3	0.1
avoid-obstacles	softmax(q)	1 − max IR − bump	0.95	0.3	0.1

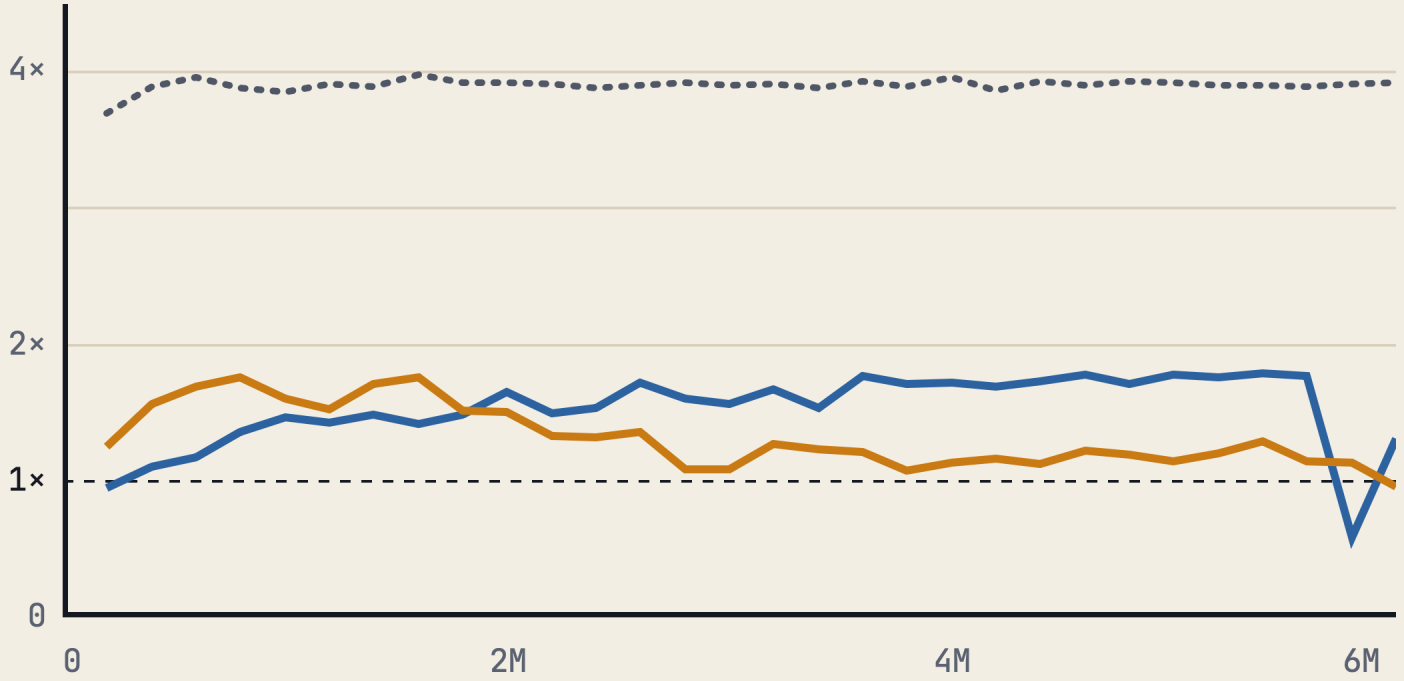
What each demon expects, and what actually happens

$\hat{q}(s, a)$ vs $G = \Sigma_k\,(\Pi\,\gamma)\,C_{t+k+1}$, rolled out with π



09 · FIRST RUN, 0 – 6M STEPS

Avoid and spin learn; seek-light fades



learned ÷ random reward per step, 200k-step means of evaluations every 25k steps

3.85×
spin at 6M

1.51×
avoid; briefly −0.39 vs 0.42 at 5.7M

0.92×
seek-light; below random in 37 of 240 evaluations

Seek-light's reward barely changed with heading

before · all four sensors

$$\Sigma_j L_j \in [1, \sqrt{2}] \cdot b(d)$$

for every heading: 0.30 – 0.45 at (2.5, 2.5)

after · front two only

$$L_\mathrm{FL} + L_\mathrm{FR} \in [0, \sqrt{2}] \cdot b(d)$$

≈ 0 – 0.46 at the same spot, same falloff: turning toward the light now pays

Learned action choice at 6M steps, 200 starts (seek_check.py)

light clearly on the left



FWD 38 · BACK 45 · LEFT 11 · RIGHT 2 · STOP 4 (%)

light clearly on the right



FWD 25 · BACK 54 · LEFT 14 · RIGHT 1 · STOP 6 (%)

With the light on its right it turned **left** 14× more often than right. It reached the light from 15% of starts; random reached it from 34%.

How we choose α,λ

seek-light · learned ÷ random at 1M

α \ λ	0	0.3	0.6	0.9
0.01	1.32	1.33	1.35	1.10
0.03	1.29	1.37	1.25	0.70
0.1	3.61	4.53	1.04	0.24
0.3	4.78	1.48	0.24	0.24

avoid-obstacles · learned ÷ random at 1M

α \ λ	0	0.3	0.6	0.9
0.01	0.93	1.02	1.23	0.23
0.03	1.34	1.53	0.50	0.08
0.1	1.70	1.74	0.31	-1.74
0.3	1.83	0.34	-1.28	-1.74

Dashed: old setting (α 0.03, λ 0.3). Solid: chosen (α 0.1, λ 0.3). Grey: no better than random; most of these diverged. λ = 0.9 blew up at every α. Seek-light at α 0.1 peaked at 6.33× after 500k steps.

× 7.1

One rare action can multiply a demon's eligibility trace by seven.

$\gamma\lambda\rho = 0.95 \times 0.3 \times 25 \approx 7.1$ · twice in a row $\approx 50\times$

Capping ρ inside the trace stopped the blow-ups

before · GQ(λ) trace

$$e_t = \varphi_t + \gamma\lambda\rho_t\, e_{t-1} \quad \text{factor} \le 7.1$$

after · Retrace-style cap (Munos et al., 2016)

$$e_t = \varphi_t + \gamma\lambda\, \min(1, \rho_t)\, e_{t-1} \quad \text{factor} \le 0.285$$

The cap changes GQ(λ)'s trace, so its convergence proof no longer applies as stated. It is a practical fix, watched through the weight sizes.

20k steps	max γλρ	γλρ > 1	> 3 twice
seek-light	7.12	5.0 %	10 times
avoid	7.12	3.3 %	11 times
spin	7.11	2.9 %	0 times

1.3×10^7

seek-light's largest weight at 9.0M (5.3 at 6M)

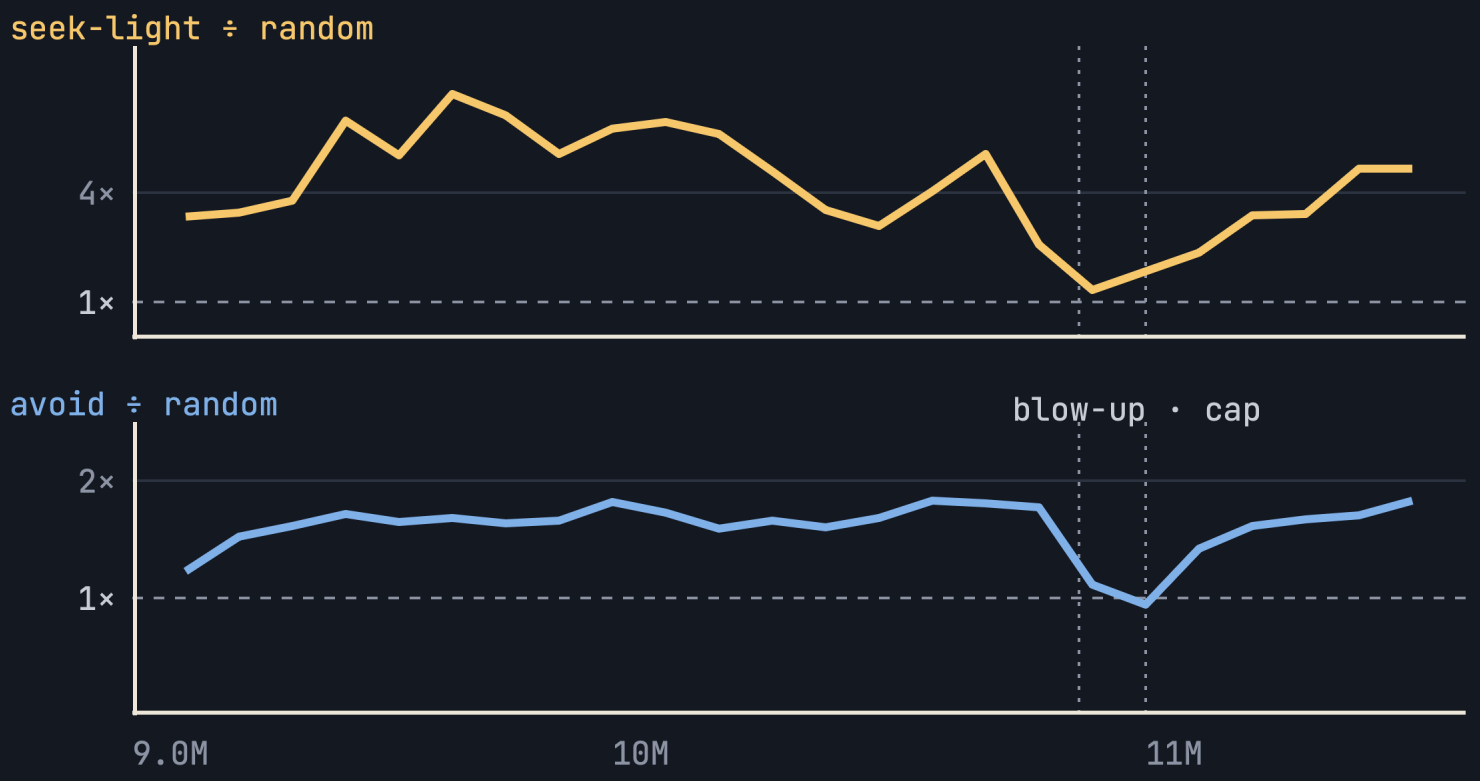
$2.5 \rightarrow 40$

avoid, within one 25k window at 10.775M

Spin never had two large factors in a row, and spin is the one control demon that never diverged.

14 · AFTER THE RESET, 9.0M – 11.4M

Fresh weights, α 0.1: seek-light near 5× random



4.80×
seek-light, mean of 58 evaluations, 9.3 – 10.75M

1.71×
avoid over the same window

1.06×
seek-light before the reset, 6 – 9M

14 · Experiments · 100k-step means · dotted lines: avoid blew up at 10.775M; ρ capped and both reset at 10.9M

15 · FINAL RUN

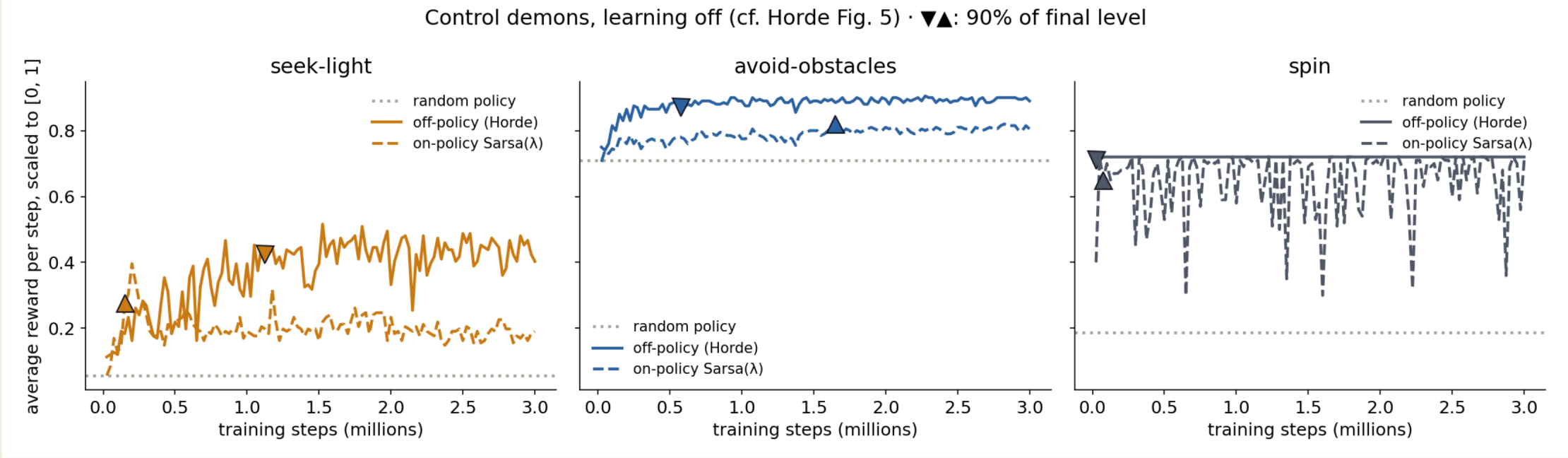
Same features, same 100 starts, three learners

demon	random	Horde, off-policy · 3M	on-policy plain · 3M	on-policy episodic · 3M
seek-light	0.08	0.57 · 7.2× 90% at 1.13M	0.27 · 3.4× 90% at 150k	1.07 · 13.5× 90% at 425k
avoid-obstacles	0.42	0.78 · 1.9× 90% at 575k	0.61 · 1.5× 90% at 1.65M	0.82 · 2.0× 90% at 325k
spin	0.19	0.72 · 3.9× 90% at 25k	0.71 · 3.8× 90% at 75k	0.71 · 3.8× 90% at 25k

Reward per step of the greedy policy at the last evaluation, learning off, 250 steps from each start. “90% at”: first step where the trailing 100k mean covers 90% of the gap from random to that learner’s own final level.

15 · Final run · env frozen · GQ(λ) α 0.1, λ 0.3 · Sarsa(λ) λ 0.9, ε 0.1, α 0.05 · tests/final_offpolicy_vs_onpolicy

Off-policy ends higher on all three demons



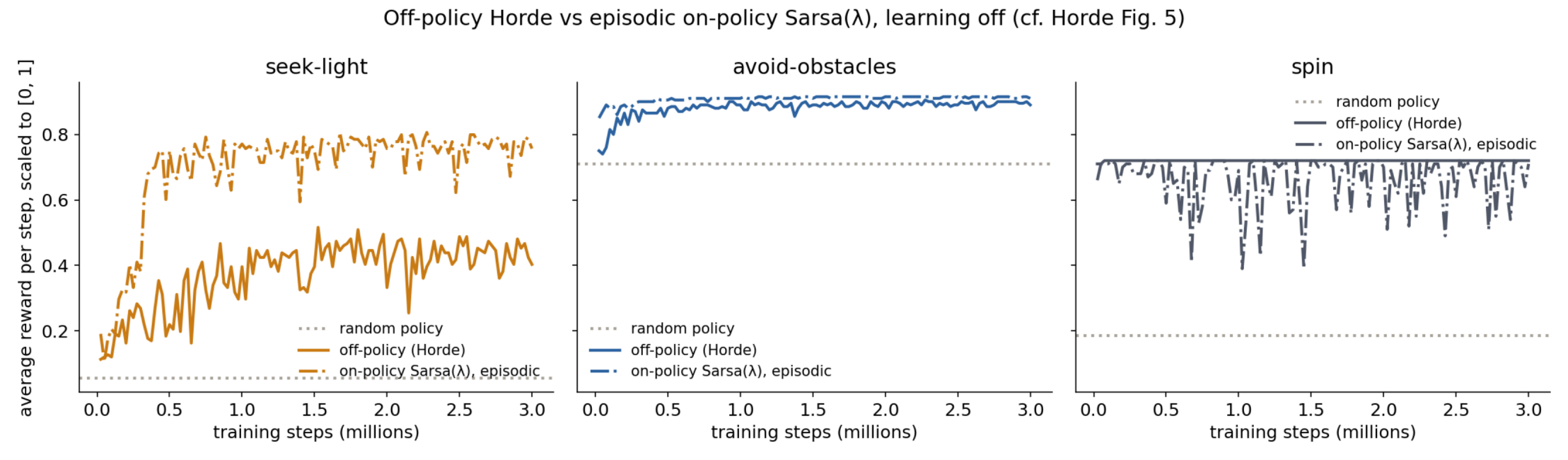
0.44 vs 0.18
seek-light

0.90 vs 0.81
avoid-obstacles

0.72 vs 0.62
spin

97.5%
on-policy steps parked near the light

With restarts, on-policy wins seek-light



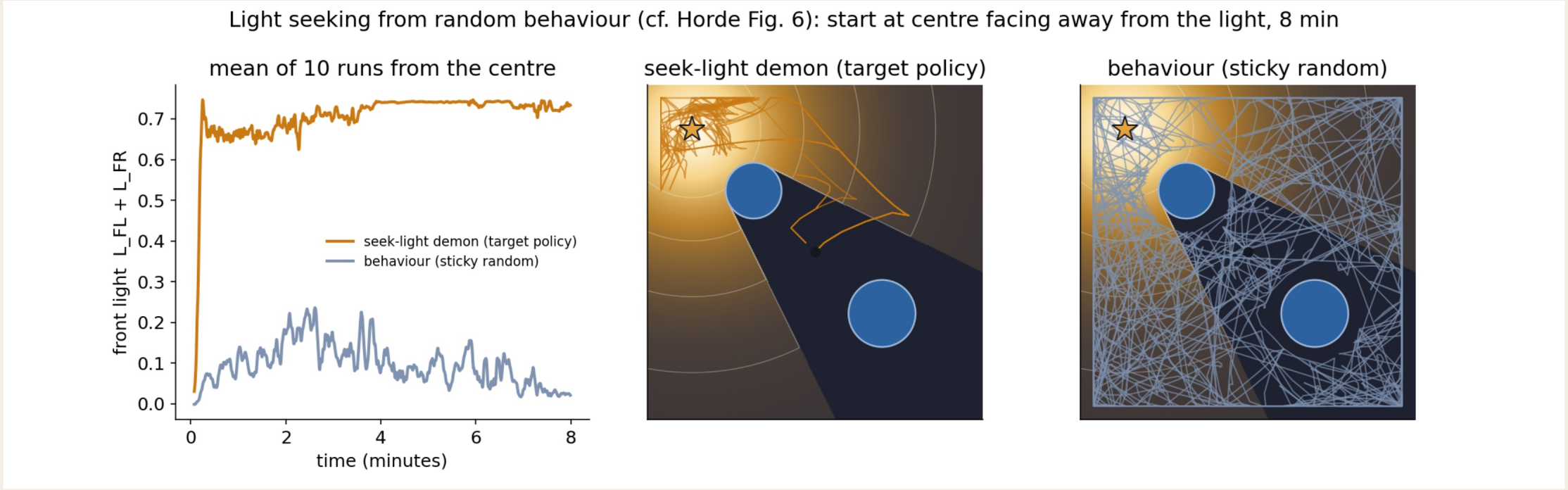
0.76 vs 0.44
seek-light, episodic vs Horde

0.91 vs 0.90
avoid-obstacles

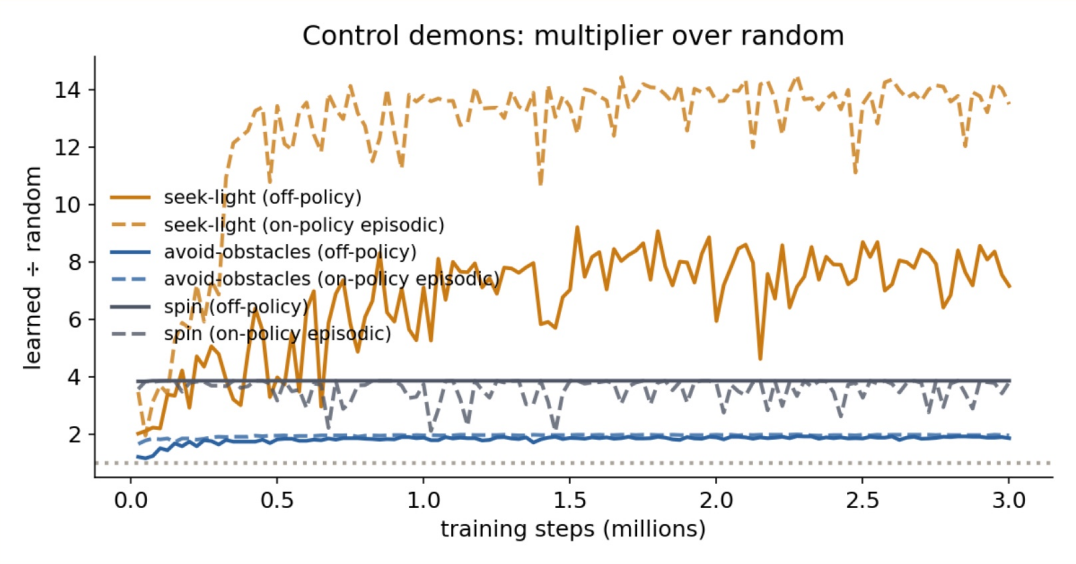
0.67 vs 0.72
spin

1 vs 7
demons learned per run

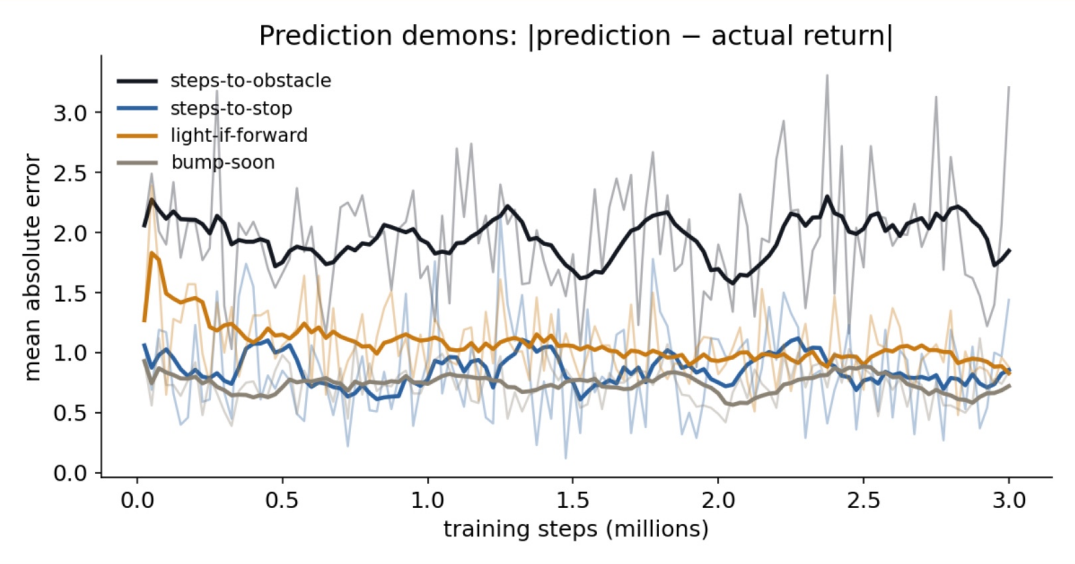
The demon finds the light its teacher never sought



Multipliers and prediction errors



learned ÷ random · solid Horde, dashed episodic on-policy



mean |q̂ − G| over 25 visited states · bold = 8-evaluation trailing mean

Adding values: w = 0.25 cuts bumps tenfold

$$a = \text{argmax}_a \left[q_\text{seek} + w \cdot q_\text{avoid} \right]$$

policy	light	bumps	end
random	0.078	80.8	8%
seek	0.595	424.5	78%
avoid	0.044	2.0	0%
w 0.25	0.659	41.9	84%
w 0.5	0.569	29.8	62%
w 1.0	0.297	1.6	3%
on-pol. epi.	1.086	82.1	92%

20 · Using the knowledge · Horde 3M · 100 shared starts × 250 steps · bumps per 1k steps · end = ended within 0.5 m of the light

21 · POLICY VARIANCE

Why seek-light's score jumps between evaluations

50%

of visited states get a different best action after 25k more steps

0.088

median gap between the best and second-best action's value

0.358

mean change of q over those 25k steps: four times the gap

±0.02

score spread from sensor noise alone, weights frozen

A constant α keeps q moving by more than the margins that decide the action, so the greedy policy flips. Planned fix: act on Polyak-averaged weights.

$$\bar{\theta} \leftarrow 0.999\,\bar{\theta} + 0.001\,\theta$$

New tasks with no new learning: successor features + GPI

$$Q_{i,w}(s,a) = \Sigma_j \; w_j \; \psi_{i,j}(s,a)$$

$$\pi(s) \in \text{argmax}_a \max_i Q_{i,w}(s,a)$$

$$Q\pi \geq \max_i Q_i - 2\varepsilon / (1-\gamma)$$

$\psi_{i,j}$ is just a GVF: cumulant j, following control demon i's policy. Linearity gives any reward $\Sigma_j \; w_j \; c_j$ for free; the bound is Theorem 1 of Barreto et al. (2017).

`light - 2·bump` · seek safely

`- light` · hide in the shadow (expected weak)

`speed + avoid` · wander safely

ψ grid · policy × cumulant

	light	avoid	bump	speed
seek	exists	new	new	new
spin	new	new	new	new
avoid	new	exists	new	new

10 new demons, 7 → 17 in the Horde. One line in horde.py lets a demon follow another demon's policy.

What the experiments taught us

7.2×

off-policy seek-light at 3M, learned from random driving alongside six other demons

3.4× · 13.5×

on-policy Sarsa(λ) at 3M: plain parks at the light, random restarts fix it

424 → 42

bumps per 1k steps when seek adds 0.25 × avoid, with more light, not less

7.1 → 0.285

largest one-step trace factor, before and after capping ρ at 1; no divergence since

next · GPI composition · Polyak-averaged policies · moving obstacles